



Semester II 2020/2021

Subject : Bioinformatics I (SCSB2103)
Section : 01 – Dr Haslina Hashim
Topic : Lab 03 – Pairwise Sequence Alignment

KOO XUAN (A23CS0300)

1) Perform pairwise alignment at the NCBI BLAST website.

- Obtain the human HBA and HBB protein sequences.
- Perform pairwise alignment at the NCBI BLAST website.
- Then use a comparison tool from the EBI website.
- Vary the scoring matrix (e.g., try different PAM and BLOSUM matrices) and record the effects on the score, the number of gaps, the percent identity, and the length of the aligned region.
- For the NCBI BLASTP program note that the output of a pairwise alignment includes a dot matrix view.

HBA:

The screenshot displays the NCBI Gene database interface for the gene HBA1 (hemoglobin subunit alpha 1). The main content area shows the gene name, its accession number (NM_000558.5), and a table of RefSeq sequences. The table has columns for Transcript, nt, Protein, aa, Isoform, and Status. The sequence NM_000558.5 is listed with a protein length of 142 amino acids. The right sidebar contains filters and search options, including a search bar and a 'Find related data' section. The bottom of the page shows the Windows taskbar with the time 2:54 PM on 17/4/2025.

Transcript	nt	Protein	aa	Isoform	Status
NM_000558.5	577	NP_000549.1	142	MANE Select	

From here we can find the RefSeq accession number and sequence of HBA1 protein.

They are:

>NP_000549.1 HBA1 [organism=Homo sapiens] [GeneID=3039]

MVLSPADKTNVKAAWGKVGHAHAGEYGAEALERMFLSFPTTKTYFPHFDLSHGSAQVKG
HGKKVADALTNAVAHVDDMPNALSALSDLHAHKLRVDPVNFKLLSHCLLVTLAAHLPA
EFTPAVHASLDKFLASVSTVLTSKYR

HBB:

The screenshot displays the NIH Gene database interface for the HBB gene. The header includes the NIH logo and the text 'National Library of Medicine National Center for Biotechnology Information'. The search bar shows 'Gene' and 'HBB'. The main content area displays the gene name 'HBB - hemoglobin subunit beta' and its function 'CD113L-C, ECT6, beta-globin'. Below this, there are buttons for 'RefSeq products', 'Orthologs', and 'Genome Data Viewer'. A sidebar on the right shows 'Results by taxon' with a list of organisms including Homo sapiens, Mus musculus, Bos taurus, Rattus norvegicus, and Salmo salar. The bottom of the page shows search results for 'HBB' with 1444 items found.

From here we can find the RefSeq accession number and sequence of HBB protein. They are:

>NP_000509.1 HBB [organism=Homo sapiens] [GeneID=3043]

MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAMGN
PKVKAHGKKVLGAFSDGLAHLNLTGTFATLSELHCDKLHVDPENFRLLGNVLVCVLA
HHFGKEFTPPVQAAYQKVVAGVANALAHKYH

Then we align them at NCBI:

BLAST® » blastp suite Home Recent Results Saved Strategies Help

blastn **blastp** blastx tblastn tblastx **Align Sequences Protein BLAST**

BLASTP programs search protein subjects using a protein query. [more...](#) [Reset page](#) [Bookmark](#)

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#) **Query subrange** [?](#)

NP_000509.1 From
To

Or, upload file Choose File No file chosen [?](#)

Job Title [?](#)
 Enter a descriptive title for your BLAST search [?](#)

☒ Align two or more sequences [?](#)

Enter Subject Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#) **Subject subrange** [?](#)

NP_000549.1 From
To

Or, upload file Choose File No file chosen [?](#)

Program Selection

Algorithm ☒ blastp (protein-protein BLAST) [Choose a BLAST algorithm](#) [?](#)

BLAST Search protein sequence using Blastp (protein-protein BLAST)

26°C Sunny Q Search ENG US 8:43 AM 18/4/2025

Here is part of the result, showing the description and the dot matrix view of the result:

BLAST® » blastp suite-2sequences » results for RID-0381C2PD114 Home Recent Results Saved Strategies Help

[< Edit Search](#) [Save Search](#) [Search Summary](#) [How to read this report?](#) [BLAST Help Videos](#) [Back to Traditional Results Page](#)

Job Title NP_000509:hemoglobin subunit beta [Homo sapiens]

RID [0381C2PD114](#) Search expires on 04-19 08:43 am [Download All](#) [?](#)

Program Blast 2 sequences [Citation](#) [?](#)

Query ID [NP_000509.1](#) (amino acid)

Query Descr hemoglobin subunit beta [Homo sapiens]

Query Length 147

Subject ID [NP_000549.1](#) (amino acid)

Subject Descr hemoglobin subunit alpha [Homo sapiens]

Subject Length 142

Other reports [Multiple alignment](#) [MSA viewer](#) [?](#)

Filter Results

Percent Identity to **E value** to **Query Coverage** to [Filter](#) [Reset](#)

Descriptions [Graphic Summary](#) [Alignments](#) [Dot Plot](#)

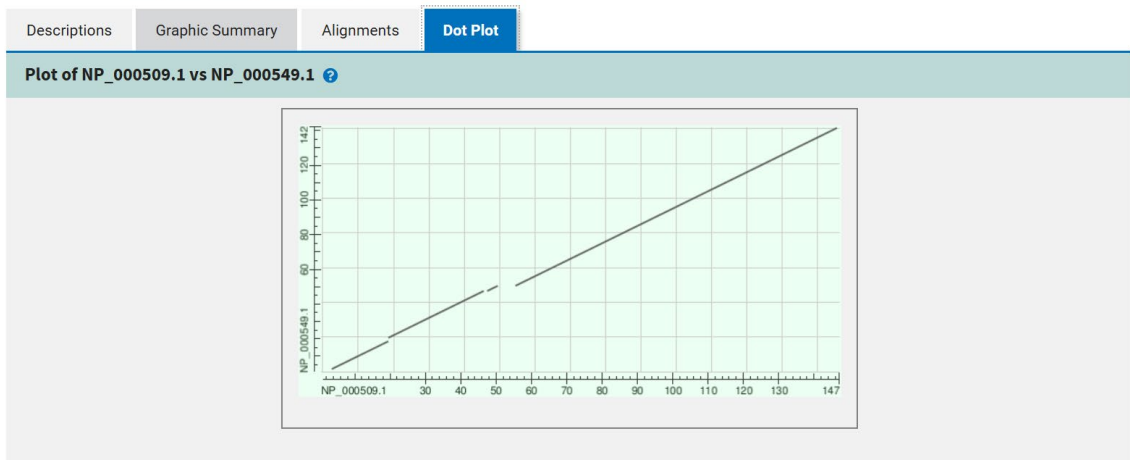
Sequences producing significant alignments [Download](#) [Select columns](#) [Show 100](#) [?](#)

☒ select all 1 sequences selected [GenPept](#) [Graphics](#) [Multiple alignment](#) [MSA Viewer](#)

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per Ident	Acc. Len	Accession
☒ hemoglobin subunit alpha [Homo sapiens]	Homo sapiens	114	114	97%	2e-38	43.45%	142	NP_000549.1

26°C Sunny Q Search ENG US 8:44 AM 18/4/2025

Subject ID [NP_000549.1](#) (amino acid)
 Subject Descr hemoglobin subunit alpha [Homo sapiens]
 Subject Length 142
 Other reports [Multiple alignment](#) [MSA viewer](#) ?



Visit EBI to try a search tool and see a list of relevant tools. Choose EMBOSS water.

→ ↻ 🔍 <https://www.ebi.ac.uk/services/data-resources-and-tools/#> ☆ ⌵ ⌵ ⌵

sequences

TOOL [Web API](#)
 Tags: Pairwise sequence alignment, Global alignment

 **EMBOSS transeq**
 Nucleotide sequence translation in selected frames
TOOL [Web API](#)
 Tags: DNA translation

 **EMBOSS water**
 Smith-Waterman local pairwise alignment of sequences
TOOL [Web API](#)
 Tags: Pairwise sequence alignment, Local alignment

 **EMDB**
 A public repository for electron microscopy (EM) density maps of macromolecular complexes and subcellular structures. EMDb covers a variety of techniques, including single-particle analysis, electron tomography, and electron (2D) crystallography.
DATA RESOURCE [Web API](#) | [CC0](#)

26°C Sunny 8:51 AM 18/4/2025

Enter the two protein sequences in the FASTA format. Many tools do not accept RefSeq identifiers and therefore it is helpful to have the FASTA formatted sequences available.

EMBOSS Water
Pairwise Sequence Alignment (PSA)

Job Dispatcher Help & Privacy Your Jobs Input form Feedback

Welcome to the Job Dispatcher website! If you need assistance or have feedback, please [contact us](#).

EMBOSS Water uses the Smith-Waterman algorithm (modified for speed enhancements) to calculate the local alignment of two sequences.

Input sequence ①

Sequence type
☒ Protein ☐ DNA

Paste your first sequence here - or use the example sequence

>NP_000549.1 HBA1 [organism=Homo sapiens] [GeneID=3043]
 MYLSPADKTNVKAAGVKGAGAGEYGAELRMFLSPFTTKYFFPHFDLSHGEAQVGHGKGVADALTNAAVHYDDMPNALSALSDLAHKLRVCPNFKLLSHCLVTLAAHLPAEFTPAVHASLDKFLASVSTVLSKYR

Choose File No file chosen

Paste your second sequence here - or use the example sequence

>NP_000509.1 HBB [organism=Homo sapiens] [GeneID=3043]
 MYHLTPKSKNTALWGKVNVDVGGEALGRLLVYPWTQRFEPGDLSTPDVAMGNPKVKAGHKIKVLGAFSDGLHLNLDKGTATLSEHCOKLHYDPENFRLLGNVLCVLAHHPGKEFTPPVQAAYQVQVAGVANALAHKYH

Choose File No file chosen Use the example Clear sequence More example inputs

Parameters

OUTPUT FORMAT ①
 pair

[More options](#)

Submit

Title
 EMBOSS Water's job

Submit

After submit, it results display:

```
#####
# Program: water
# Rundate: Fri 18 Apr 2025 01:54:35
# Commandline: water
#   -auto
#   -stdout
#   -asequence emboss_water-I20250418-015429-0554-62696334-p1m.asequence
#   -bsequence emboss_water-I20250418-015429-0554-62696334-p1m.bsequence
#   -datafile EBLOSUM62
#   -gapopen 10.0
#   -gapextend 0.5
#   -aformat3 pair
#   -sprotein1
#   -sprotein2
# Align_format: pair
# Report_file: stdout
#####

#=====
#
# Aligned_sequences: 2
# 1: NP_000549.1
# 2: NP_000509.1
# Matrix: EBLOSUM62
# Gap_penalty: 10.0
# Extend_penalty: 0.5
#
# Length: 145
# Identity:      63/145 (43.4%)
# Similarity:    88/145 (60.7%)
# Gaps:          8/145 ( 5.5%)
# Score: 293.5
#
#
#=====
```


BLOSUM62

BLAST

Search protein sequence using Blastp (protein-protein BLAST)

☐ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with + sign

Algorithm parameters

Restore default search parameters

General Parameters

Max target sequences100

Short queriesAutomatically adjust parameters for short input sequences

Expect threshold+ 10

Word size3

Max matches in a query range0

Scoring Parameters

MatrixBLOSUM62

Gap CostsExistence: 11 Extension: 1

Compositional adjustmentsConditional compositional score matrix adjustment

Filters and Masking

Filter☐ Low complexity regions

Mask☐ Mask for lookup table only☐ Mask lower case letters

BLAST

Search protein sequence using Blastp (protein-protein BLAST)

☐ Show results in a new window

Descriptions

Graphic Summary

Alignments

Dot Plot

Alignment viewPairwise

Restore defaults

Download

1 sequences selected

Download

GenPept

Graphics

Next

Previous

Descriptions

hemoglobin subunit alpha [Homo sapiens]

Sequence ID: NP_000549.1 Length: 142 Number of Matches: 1

See 59 more title(s) See all Identical Proteins(IPG)

Range 1: 3 to 141

GenPept

Graphics

Next Match

Previous Match

Score	Expect	Method	Identities	Positives	Gaps
114 bits(286)	2e-38	Compositional matrix adjust.	63/145(43%)	88/145(60%)	8/145(5%)
Query 4	LTPEEKSAVTALMGKV--NDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAMGNPKV	61	L+P +K+ V A NGKV + E G EAL R+ + +P T+ +F F DLS G+ +V		
Sbjct 3	LSPADKTNWKAAMGVGAHAGEYGAELERMFLEPTTTKTYFPHF-DLS-----HGSQV	56			
Query 62	KAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCKDKLVDDPENFRLLGWLVCVLAHFFGK	121	K HGKKV A ++ +AH+D++ + LS+LH KL VDP NF+LL + L+ LA H		
Sbjct 57	KGHGKVDADLTNAVAHVDDMPNALSDLHAHKLKRDVPNFKLSHCLLVTLAHLPA	116			
Query 122	EFTPPVQAAYQKVVAGVANAHAHKY	146	EFTP V A+ K +A V+ L L KY		
Sbjct 117	EFTPAVHASLDKFLASVSTVLSKY	141			

Related Information

Gene - associated gene details

AlphaFold Structure - 3D structure displays

Genome Data Viewer - aligned genomic context

Identical Proteins - Identical proteins to NP_000508.1

BLOSUM90

Algorithm parameters

Restore default search parameters

General Parameters

Max target sequences100

Short queriesAutomatically adjust parameters for short input sequences

Expect threshold+ 10

Word size3

Max matches in a query range0

Scoring Parameters

MatrixBLOSUM90

Gap CostsExistence: 10 Extension: 1

Compositional adjustmentsConditional compositional score matrix adjustment

Filters and Masking

Filter☐ Low complexity regions

Mask☐ Mask for lookup table only☐ Mask lower case letters

BLAST

Search protein sequence using Blastp (protein-protein BLAST)

☐ Show results in a new window

Alignment view Pairwise Restore defaults Download

1 sequences selected

[Download](#) [GenPept](#) [Graphics](#) Next Previous Descriptions

hemoglobin subunit alpha [Homo sapiens]
 Sequence ID: [NP_000549.1](#) Length: 142 Number of Matches: 1
[See 59 more title\(s\)](#) [See all Identical Proteins \(IPG\)](#)

Range 1: 3 to 141 [GenPept](#) [Graphics](#) Next Match Previous Match

Score	Expect	Method	Identities	Positives	Gaps
118 bits(275)	4e-37	Compositional matrix adjust.	63/145(43%)	81/145(55%)	8/145(5%)
Query 4	LTPPEKSAVTALMGKV--NVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAMGNPKV	61			
Sbjct 3	LSPADKTNVKAAMGKVGAGHAGEYGAEALERMFLSFPTTKTYFPHF-DLSH-----GSAQV	56			
Query 62	KAHGKKVLGAFSDGLAHLNKLGTFTLSELHCDKLHVDPENFRLLGNVLCVLAHHFGK	121			
Sbjct 57	K HGKKV A ++ AH D++ + LS+LH KL VDP NF+LL L LA H	116			
Query 122	EFTPPVQAAYQKVVAGVANALAHKY	146			
Sbjct 117	EFTP V+A K A V+ L KY	141			

Related Information
[Gene](#) - associated gene details
[AlphaFold Structure](#) - 3D structure displays
[Genome Data Viewer](#) - aligned genomic context
[Identical Proteins](#) - Identical proteins to NP_000508.1

PAM30

[Restore default search parameters](#)

General Parameters

Max target sequences: Select the maximum number of aligned sequences to display

Short queries: ☒ Automatically adjust parameters for short input sequences

Expect threshold: ?

Word size: ?

Max matches in a query range: ?

Scoring Parameters

Matrix: ?

Gap Costs: Existence: 9 Extension: 1 ?

Compositional adjustments: ?

Filters and Masking

Filter: ☐ Low complexity regions ?

Mask: ☐ Mask for lookup table only ?
☐ Mask lower case letters ?

BLAST Search protein sequence using Blastp (protein-protein BLAST)
☐ Show results in a new window

Descriptions Graphic Summary Alignments Dot Plot

Alignment view Pairwise Restore defaults Download

1 sequences selected

[Download](#) [GenPept](#) [Graphics](#) Next Previous Descriptions

hemoglobin subunit alpha [Homo sapiens]
 Sequence ID: [NP_000549.1](#) Length: 142 Number of Matches: 1
[See 59 more title\(s\)](#) [See all Identical Proteins \(IPG\)](#)

Range 1: 1 to 142 [GenPept](#) [Graphics](#) Next Match Previous Match

Score	Expect	Method	Identities	Positives	Gaps
121 bits(278)	4e-34	Compositional matrix adjust.	65/149(44%)	74/149(49%)	9/149(6%)
Query 1	MVHLTPPEKSAVTALMGKV--NVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAMGN	58			
Sbjct 1	MV L P +K V A MGKV E G EAL R+ +P T +F F DLS G	53			
Query 59	PKVKAHGKKVLGAFSDGLAHLNKLGTFTLSELHCDKLHVDPENFRLLGNVLCVLAHH	118			
Sbjct 54	AQVKGHGKKVADALTNAVAHVDDMPNALSASDLHAHKL RVD P VNF KLL SHCLLVTLAAH	113			
Query 119	FGKEFTPPVQAAYQKVVAGVANALAHKYH	147			
Sbjct 114	EFTP V+A K A V L KY	142			

Related Information
[Gene](#) - associated gene details
[AlphaFold Structure](#) - 3D structure displays
[Genome Data Viewer](#) - aligned genomic context
[Identical Proteins](#) - Identical proteins to NP_000508.1

PAM250

Algorithm parameters

General Parameters

Max target sequences: 100

Short queries: ☒ Automatically adjust parameters for short input sequences

Expect threshold: 10

Word size: 3

Max matches in a query range: 0

Scoring Parameters

Matrix: PAM250

Gap Costs: Existence: 15 Extension: 3

Compositional adjustments: Conditional compositional score matrix adjustment

Filters and Masking

Filter: ☐ Low complexity regions

Mask: ☐ Mask for lookup table only ☐ Mask lower case letters

BLAST

Search protein sequence using Blastp (protein-protein BLAST)

☐ Show results in a new window

Descriptions Graphic Summary **Alignments** Dot Plot

Alignment view: Pairwise

1 sequences selected

Download GenPept Graphics

hemoglobin subunit alpha [Homo sapiens]

Sequence ID: NP_000549.1 Length: 142 Number of Matches: 1

See 59 more title(s) See all Identical Proteins(IPG)

Range 1: 3 to 142

Score	Expect	Method	Identities	Positives	Gaps
117 bits(381)	9e-41	Compositional matrix adjust.	63/146(43%)	109/146(74%)	8/146(5%)

Query 4 LTPEEKSAVTALWGKV--NVDEVGGEALGRLLVYPWTQRRFFESFGDLSTPDAMVGNPKV 61

L+P +K+ V A WGKV + +E G+EAL R+++ +P T+ +F F DLS G+++V

Sbjct 3 LSPADKTNVKAAMGKVGAGHAGEYGAELERMF LSFPTTKTYFPHF-DLSH----GSAQV 56

Query 62 KAHGKKVLGAFSDGLAHLNKLKGTFTLSELHCDKLHVDPENFRLLGNVLCVLAHIFGK 121

K+HGKKV +A++++AH+D++ ++++LS+LH +KL+VDP NF+LL++ L+ LA H+

Sbjct 57 KGHGKKVADALTNVAHVDDMPNALSDLSLHAHLKLRVDPVNFKLLSHCLLVTLAAHLPA 116

Query 122 EFTPPVQAAYQKVVAGVANALAHKYH 147

EFTP+V+A+ +K +A+V+ L+ KY+

Sbjct 117 EFTPAVHASLDKFLASVSTVLTSKYR 142

Related Information

Gene - associated gene details

AlphaFold Structure - 3D structure displays

Genome Data Viewer - aligned genomic context

Identical Proteins - Identical proteins to NP_000508.1

Settings	Matrix	E value	Identities	Gaps
Default (see above)	BLOSUM62	2e-38	63/145	8/145
	BLOSUM90	4e-37	63/145	8/145
	PAM30	4e-34	65/149	9/149
	PAM250	9e-41	63/146	8/146

Using these other scoring matrices has a minimal effect on the results. It is often easier to see large differences when two proteins are more distantly related, or when only a portion of two proteins aligns well.

2) Perform pairwise alignment at the UCSC website.

- i) Go to <http://genome.ucsc.edu>. Follow the link to the genome browser, select the human genome hg19 build, and enter a query of hbb. This should direct you to chr11:5,246,696-5,248,301 (a region of 1,606 base pairs encompassing the beta globin gene, HBB).
- ii) Click the box to set the view to default tracks.
- iii) Under Comparative Genomics” select Placental Chain/Net and set the display to full. By clicking the Placental Chain/Net header you can view a series of options. Set Chains to full view and Nets to full view. Set the species to horse (deselect other species). Click submit.
- iv) The display now shows human/horse chained alignments and alignment nets.

Enter hbb at UCSC Genome Browser

The screenshot shows the UCSC Genome Browser homepage. The header features the UCSC logo and navigation links: Genomes, Genome Browser, Tools, Mirrors, Downloads, My Data, Projects, Help, About Us. A search bar with the text 'hbb' and a 'Search' button is prominently displayed. Below the search bar, there are two main sections: 'Tools' and 'News'. The 'Tools' section lists various genomic tools with icons and brief descriptions: Genome Browser (Interactively visualize genomic data), BLAT (Rapidly align sequences to the genome), In-Silico PCR (Rapidly align PCR primer pairs to the genome), Table Browser (Download and filter data from the Genome Browser), LiftOver (Convert genome coordinates between assemblies), REST API (Returns data requested in JSON format), and Variant Annotation Integrator (Annotate genomic variants). The 'News' section lists recent updates and publications, including 'New Pseudogenes track for hg38', 'denovo-db tracks for hg19 now available', 'New MITOMAP track for hg38 and hg19', 'New Splicing Impact super track and SpliceVarDB track for hg38', 'New highlight features for track hubs now available', and 'enGenome VarChat track for human (hg38 and hg19)'. At the bottom, there are banners for 'Meetings and Workshops: Come see us in person!' including 'CSHL Biology of Genomes -- Cold Spring Harbor, NY. May 6-10, 2025' and 'ESHG: European Human Genetics -- Milan, Italy. May 24-27, 2025. Visit us at booth 2020!'. The footer shows the current date and time: 10:26 AM 18/4/2025.

Then click HBB:

UCSC Genomes Genome Browser Tools Mirrors Downloads My Data Projects Help About Us

Search across the Genome Browser site

Search:

Human Feb. 2009 (GRCh37/hg19)

Use the tree to hide/show results from only these categories. Hover your mouse over each category for an explanation:

- hg19 Track Data (674 results)
 - Visible Tracks (17 results)
 - HGNC (3 results)
 - RefSeq Curated (2 results)
 - GENCODE V47/Inf37 (12 results)
 - Currently Hidden Tracks (657 results)
 - Mapping and Sequencing (1 results)
 - STS Markers (1 results)
 - Genes and Gene Predictions (179 results)
 - Plam in UCSC Gene (1 results)
 - Other RefSeq (49 results)
 - Old UCSC Genes (3 results)
 - IKMC Genes Mapped (2 results)
 - Ensembl Genes (7 results)
 - NCBI RefSeq (3 results)
 - Vega Genes (2 results)
 - GENCODE Versions (6 results)
 - Phenotypes, Variants, and Literature (1 results)
 - GeneReviews (1 results)
 - mRNA and EST (476 results)

Search Results on hg19 (Human Feb. 2009 (GRCh37/hg19))

- ☐ HUGO Gene Nomenclature:
 - HBB** - chr11:5246695-5248301
 - HBBP1** - chr11:5263185-5264822
 - SCN5A** - chr3:38589554-38691178
- ☐ Gencode Genes:
 - HBB** (ENST00000335295.4_8) - chr11:5246694-5248301 - hemoglobin subunit beta (from RefSeq NM_000518.5)
 - HBD** (ENST00000650601.1_9) - chr11:5254068-5255713 - hemoglobin subunit delta (from RefSeq NM_000519.4)
 - RBM17** (ENST00000379888.9_4) - chr10:6130997-6159410 - RNA binding motif protein 17, transcript variant 1 (from RefSeq NM_032905.5)
 - HBM** (ENST00000356815.4_4) - chr16:215977-216763 - hemoglobin subunit mu (from RefSeq NM_001003938.4)
 - GATA1** (ENST00000376670.9_10) - chrX:4864498-48652718 - GATA binding protein 1 (from RefSeq NM_002049.4)
 - HBQ1** (ENST00000199708.3_4) - chr16:230458-231178 - hemoglobin subunit theta 1 (from RefSeq NM_005331.5)
 - KLF2** (ENST00000248071.6_4) - chr19:16435637-16439496 - Kruppel like factor 2 (from RefSeq NM_016270.4)
 - HBZ** (ENST00000252951.3_4) - chr16:202643-204502 - hemoglobin subunit zeta (from RefSeq NM_005332.3)
 - HBA1** (ENST00000320868.9_4) - chr16:226679-227521 - hemoglobin subunit alpha 1 (from RefSeq NM_000558.5)
 - HBA2** (ENST00000251595.11_5) - chr16:222875-223709 - hemoglobin subunit alpha 2 (from RefSeq NM_000517.6)
- ☐ Show 2 more matches for Gencode Genes
- ☐ NCBI RefSeq genes, curated subset (NM_*, NR_*, NP_* or YP_*):
 - NR_001589.1** - chr11:5263184-5264822
 - NM_000518.5** - chr11:5246694-5248301
- ☐ RefSeq Genes:
 - HBB** - chr11:5246694-5248301 - (NM_000518) hemoglobin subunit beta
 - HBBP1** - chr11:5263185-5264822 - (NR_001589)
- ☐ NCBI RefSeq Other Annotations (not NM_*, NR_*, XM_*, XR_*, NP_* or YP_*):
 - KRT69P** - chr12:52738003-52746938
- ☐ Non-Human RefSeq Genes:
 - HBB** - chr11:5254179-5255722 - (NM_001279263) hemoglobin, beta
 - HBB** - chr11:5246828-5248310 - (NM_001279263) hemoglobin, beta
 - hbb** - chr11:5247810-5275746 - (NM_001201019) hemoglobin subunit beta
 - HBB** - chr11:5246696-5248295 - (NM_001329918) hemoglobin subunit beta
 - HBB** - chr11:5246718-5248301 - (NM_001283367) hemoglobin subunit beta
 - Hbb** - chr11:5246696-5255672 - (NM_033234) hemoglobin subunit beta-1

30°C Mostly sunny

The Comparative Genomics options are here and click on placental chain/net full view:

gPCR Primers

RIKEN CAGE Loc Seston Brain

Regulation

ENCORE Regulation	CD34 DnaseI	CpG Islands	ENC Chromatin	ENC DNA Methyl	ENC DNase/FAIRE	ENC Histone	ENC RNA Binding
ENC TF Binding	FANTOM5	FSU Repli-chip	GeneHancer	Genome Segments	GTEx Combined eQTL	GTEx Tissue eQTL	JASPAR Transcription Factors
NKI Nuc Lamina	OREGAnno	Rao 2014 Hi-C	ReMap ChIP-seq	Stanf Nucleosome	SUNY SwitchGear	SwitchGear TSS	TFBS Conserved
TS miRNA Targets	UCSF Brain Methyl	UMMS Brain Hist	UW Repli-seq	VISTA Enhancers			

Comparative Genomics

Conservation	Cons 46-Way	Cons Indels MmCf	Evo Cpg	GERP	phastBias gBGC	Primate Chain/Net	Placental Chain/Net
Vertebrate Chain/Net	CHM13 alignments						

Neandertal Assembly and Analysis

Denisova Assembly and Analysis

Repeats

RepeatMasker	Interrupted Repts	Microsatellite	NumtS Sequence	Segmental Dups	Self Alignment	Simple Repeats	WM + SDust
dense	hide	hide	hide	hide	hide	hide	hide

z://genome.ucsc.edu/cgi-bin/hgTrackUi?hgsid=2520915545_ZRtWAS5sRwvnmDLPTUgh5iaAW8db=hg19&c=chr11&g=placentalChainNet

ly sunny

Click the header of that track to view more options; deselect all (-key) then select horse, and set chains and nets from pack or dense to full:

Placental Chain/Net Track Settings [Subtracks](#) [Description](#)

Non-primate Placental Mammal Genomes, Chain and Net Alignments (All Comparative Genomics tracks)

Maximum display mode: [Reset to defaults](#)

Select views ([Help](#)):
[Chains](#) [Nets](#)

Select subtracks by clade and species:

Species	Euarchontoglires	Carnivora	Cetartiodactyla	Laurasiatheria	Perissodactyla	Afrotheria
Guinea pig	☐					
Chinese hamster	☐					
Mouse	☐					
Pika	☐					
Rabbit	☐					
Rat	☐					
Panda		☐				
Dog		☐				
Cat		☐				
Cow			☐			
Sheep			☐			
Pig			☐			
Alpaca			☐			
Hedgehog				☐		
Shrew				☐		
White rhinoceros					☐	
Horse					☒	
Tenrec						☐
Elephant						☐

0°C
fostly sunny

10:34 AM
18/4/2025

Click Submit. The resulting Genome Browser display shows chained alignments and alignment nets.



Click an alignment block to view the pairwise alignment between human and horse DNA at that region.

3) Using the amino acid explorer tool from NCBI

- i) Visit http://www.ncbi.nlm.nih.gov/Class/Structure/aa/aa_explorer.cgi Select the Biochemical Properties table. Which amino acid is most abundant? (Is it leucine, at 9.94%?). Use this table to test yourself and make sure you know the one- and three-letter abbreviations for all 20 amino acids, as well as their structures.
- ii) Is tyrosine a hydrophobic amino acid? To decide, use the Common Substitutions table. Explore valine (a hydrophobic residue), sort the results by hydrophobicity, and see where tyrosine is located. You can also explore the Structure and Chemistry table.

The website doesn't exist

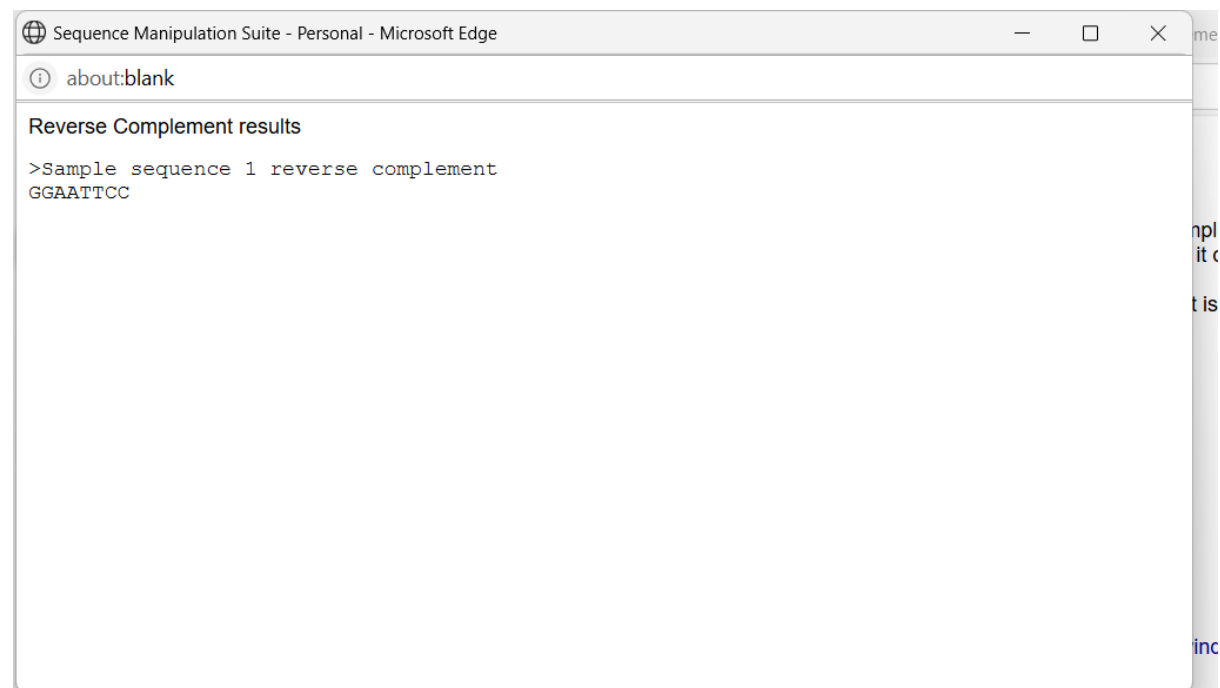
4) Sequence Manipulation Suite

- i) Many tools are available to manipulate sequences. Visit the Sequence Manipulation Suite (<http://www.bioinformatics.org/sms2/index.html>) to access a large number of tools. What is the reverse complement of the sequence GGAATTCC?

Visit website <http://www.bioinformatics.org/sms2/index.html> , then click reverse complement and enter sequence GGAATTCC in the enter box and click submit.

The screenshot shows the 'Sequence Manipulation Suite' website. The 'Reverse Complement' tool is selected. The input field contains the sequence 'GGAATTCC'. The 'Submit' button is visible. The website has a sidebar with various tools categorized into 'Format Conversion', 'Sequence Analysis', 'Sequence Figures', and 'Random Sequences'. The main content area explains the 'Reverse Complement' tool and provides a text box for the sequence. The browser's taskbar at the bottom shows the date and time as 10:46 AM on 18/4/2025.

The results show:



5) Perform pairwise alignment using EMBOSS tools via Galaxy and UCSC

- i) In this exercise we will perform global alignment with the EMBOSS package needle and local alignment with the EMBOSS package water.
- ii) Both of these are available at the Galaxy public web server (along with over 100 other EMBOSS tools). Box 3.9 introduces EMBOSS and explains how to import beta globin (HBB) and alpha globin (HBA2) proteins from the UCSC Table Browser using Galaxy, and to then align them. This history is saved at <https://main.g2.bx.psu.edu/u/pevsner/h/pairwise-alignment-via-ucsc-and-emboss> (WebLink 3.14).
- iii) Note that Galaxy is a web-based platform for using hundreds of bioinformatics tools, including next-generation sequence data analysis software.
- iv) To use it visit <http://usegalaxy.org> then go to the public server.
- v) Be sure to create a username and log in. This will allow you to continue your work over time and at different work stations.

First, visit galaxy website and log in, and then go to UCSC Main table browser

The screenshot shows the Galaxy web interface. On the left sidebar, under 'Tools', the 'UCSC Main table browser' is selected. The main content area displays a welcome message and a banner for GBCC2025. The right sidebar shows an empty 'History' section. The browser's address bar shows the URL: https://usegalaxy.org/tool_runner/data_source_redirect?tool_id=ucsc_table_direct1. The system tray at the bottom indicates the temperature is 7°C and it is 'irrtly sunny'.

Then enter HBA2 at the position and select FASTA output format and get output

The screenshot shows the UCSC Table Browser interface. The 'Select dataset' section has 'Clade' set to 'Mammal', 'Genome' set to 'Human', and 'Assembly' set to 'Feb. 2009 (GRCh37/hg19)'. The 'Group' is 'Genes and Gene Predictions', 'Track' is 'NCBI RefSeq', and 'Table' is 'RefSeq All (ncbiRefSeq)'. The 'Define region of interest' section has 'Region' set to 'Position' with the value 'chr16:222,876-223,709'. The 'Optional: Subset, combine, compare with another track' section has 'Filter' set to 'Create', 'Subtrack merge' set to 'Create', 'Intersection' set to 'Create', and 'Correlation' set to 'Create'. The 'Retrieve and display data' section has 'Output format' set to 'CDS FASTA alignment from multiple alignment', 'Send output to' checked for 'Galaxy', and 'Output filename' set to '(leave blank to keep output in browser)'. The 'File type returned' is set to 'Plain text'. The 'Get output' button is visible. The browser's address bar shows the URL: <https://ucscbioinformatics.genome.ucsc.edu/tableBrowser/>. The system tray at the bottom indicates the temperature is 30°C and it is 'Partly sunny'.

Deselect all species selection and select horse and click submit

Select multiple alignment and options for NCBI RefSeq

MAF table: multiz100way

Formatting options:

- ☐ Separate into exons
- ☐ Show nucleotides
- ☐ Output lines with just dashes
- ☐ Format output as table ☐ Truncate headers at characters (enter zero for no headers)

Species selection: Defaults

Primate

- ☐ chimp ☐ gorilla ☐ orangutan ☐ gibbon ☐ rhesus
- ☐ crab-eating macaque ☐ baboon ☐ green monkey ☐ marmoset ☐ squirrel monkey
- ☐ bushbaby

Euarchontoglires

- ☐ chinese tree shrew ☐ squirrel ☐ lesser Egyptian jerboa ☐ prairie vole ☐ chinese hamster
- ☐ golden hamster ☐ mouse ☐ rat ☐ naked mole-rat ☐ guinea pig
- ☐ chinchilla ☐ brush-tailed rat ☐ rabbit ☐ pika

Laurasiatheria

- ☐ pig ☐ alpaca ☐ bactrian camel ☐ dolphin ☐ killer whale
- ☐ tibetan antelope ☐ cow ☐ sheep ☐ domestic goat ☒ horse
- ☐ white rhinoceros ☐ cat ☐ dog ☐ ferret ☐ panda
- ☐ pacific walrus ☐ weddell seal ☐ black flying-fox ☐ megabat ☐ david's myotis (bat)
- ☐ little brown bat ☐ big brown bat ☐ hedgehog ☐ shrew ☐ star-nosed mole

Afrotheria

Click send query to galaxy

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Output results to Galaxy as NCBI RefSeq

Repeat again and put HBB for the position:

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Table Browser

Use this tool to retrieve and export data from the Genome Browser annotation track database. You can limit retrieval based on data attributes and intersect or merge with data from another track, or retrieve DNA sequence covered by a track. [More...](#)

Select dataset

Clade: Mammal Genome: Human Assembly: Feb. 2009 (GRCh37/hg19)

Group: Genes and Gene Predictions Track: NCBI RefSeq

Table: RefSeq All (ncbiRefSeq)

Define region of interest

Region: ☐ Genome ☐ ENCODE Pilot regions ☒ Position

Identifiers (names/accessions):

Optional: Subset, combine, compare with another track

Filter:

Subtrack merge:

Intersection:

Correlation:

Retrieve and display data

Output format: CDS FASTA alignment from multiple alignment ☒ Send output to ☒ Galaxy ☐ GREAT

Output filename: (leave blank to keep output in browser)

File type returned: ☒ Plain text ☐ Gzip compressed

Using the Table Browser

This section provides brief help by listing descriptions of the Table Browser controls. For more information on using this program, see the Table Browser User's Guide.

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Table Browser

Your search resulted in multiple matches. Please select a position:

HUGO Gene Nomenclature

[HBB_at_chr11:5246695-5248301](#)
[HBBP1_at_chr11:5263185-5264822](#)
[SCN5A_at_chr3:38589554-38691178](#)

[More results...](#)

Deselect all species selection and select horse and click submit

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Select multiple alignment and options for NCBI RefSeq

MAF table: [multiz100way](#)

Formatting options:

- ☐ Separate into exons
- ☐ Show nucleotides
- ☐ Output lines with just dashes
- ☐ Format output as table ☐ Truncate headers at characters (enter zero for no headers)

Species selection: [+](#) [-](#) [Defaults](#)

Primate [+](#) [-](#)

- ☐ chimp ☐ gorilla ☐ orangutan ☐ gibbon ☐ rhesus
- ☐ crab-eating macaque ☐ baboon ☐ green monkey ☐ marmoset ☐ squirrel monkey
- ☐ bushbaby

Euarchontoglires [+](#) [-](#)

- ☐ chinese tree shrew ☐ squirrel ☐ lesser Egyptian jerboa ☐ prairie vole ☐ chinese hamster
- ☐ golden hamster ☐ mouse ☐ rat ☐ naked mole-rat ☐ guinea pig
- ☐ chinchilla ☐ brush-tailed rat ☐ rabbit ☐ pika

Laurasiatheria [+](#) [-](#)

- ☐ pig ☐ alpaca ☐ bactrian camel ☐ dolphin ☐ killer whale
- ☐ tibetan antelope ☐ cow ☐ sheep ☐ domestic goat ☒ horse
- ☐ white rhinoceros ☐ cat ☐ dog ☐ ferret ☐ panda
- ☐ pacific walrus ☐ weddell seal ☐ black flying-fox ☐ megabat ☐ david's myotis (bat)
- ☐ little brown bat ☐ big brown bat ☐ hedgehog ☐ shrew ☐ star-nosed mole

Afrotheria [+](#) [-](#)

10°C
littly sunny

Search

ENG US 11:05 AM 18/4/2025

Click send query to galaxy

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Output results to Galaxy as NCBI RefSeq

[Send query to Galaxy](#)
[Cancel](#)

After the query successfully add to galaxy website, search for EMBOSS water

The screenshot shows the Galaxy web interface. On the left, the 'All Tools' sidebar is open, and 'EMBOSS' is selected in the search results. The main panel displays a promotional banner for the James P. Taylor Foundation for Open Science. The right sidebar shows the 'History' tab with two unnamed history entries. The bottom status bar indicates the user is logged in as 'xuan-04' and the system is using 0% of 250.0 GB.

The screenshot shows the Galaxy web interface with the 'All Tools' sidebar open. The search results for 'water' are displayed, showing various tools like 'trimseq', 'twofeat', 'union', 'vectorstrip', 'water', 'wobble', 'wordcount', and 'wordmatch'. The main panel remains the same as the previous screenshot. The bottom status bar shows the user is logged in as 'xuan-04' and the system is using 0% of 250.0 GB.

Select the 2 results just import and run the tool

The screenshot shows the Galaxy web interface. On the left, the 'Tools' panel lists various bioinformatics tools. The 'water' tool is selected. The 'Tool Parameters' section shows two sequences being aligned. The 'Run Tool' button is at the top right of the parameters section. On the far right, the 'History' panel shows a list of datasets, with the two sequences used in the tool highlighted in green.

This is the results:

The screenshot shows the Galaxy web interface with the results of the 'water' tool. The results are displayed in a text format, showing alignment statistics and the aligned sequences. The right sidebar shows the history of datasets, including the two sequences used and the resulting alignment.

```

#####
# Program: water
# Rundate: Fri 18 Apr 2025 03:06:59
# Commandline: water
#   -asequence /corral4/main/objects/a/8/b/dataset_a8b599ae-5157-43f9-ae0c-84067cb25e21.dat
#   -bsequence /corral4/main/objects/f/6/d/dataset_f6da880b-4583-4808-8e39-197b1c8fcl1f6.dat
#   -outfile /corral4/main/jobs/066/907/66907251/outputs/dataset_3860bc43-86ff-46a9-881a-56ba3c05513d.dat
#   -gapopen 10.0
#   -gapextend 0.5
#   -brief yes
#   -aformat3 srs
#   -auto
# Align_format: srs
# Report_file: /corral4/main/jobs/066/907/66907251/outputs/dataset_3860bc43-86ff-46a9-881a-56ba3c05513d.dat
#####
#=====
#
# Aligned_sequences: 2
# 1: NM_000517.6 hg19
# 2: NM_000518.5 hg19
# Matrix: EBLOSUM62
# Gap_penalty: 10.0
# Extend_penalty: 0.5
#
# Length: 147
# Identity:      64/147 (43.5%)
# Similarity:   89/147 (60.5%)
# Gaps:         8/147 ( 5.4%)
# Score: 297.5
#
#=====
NM_000517.6_h 3 LSPADKTNVKAAMGKVGAGHAGEYGAELERMFLSPTTKTYFPFH-DLS- 50
NM_000518.5_h 4 LTPEEKSAVTALNKGKV--INDEVGEALGRLLVVPWTQRFFESFGDLST 51
NM_000517.6_h 51 ----HCSAQVKGHGKVDADLTNAVAVDDPMPNALSALSDLAHKLRVDP 96
NM_000518.5_h 52 PDVAVGNPKVKAHGKKVLGAFSDGLAHNLNLTGTFATLSLHCKDLKLVDP 101
NM_000517.6_h 97 VNFKLLSHCLLVTLAAHLPAEFPAVHASLDKFLASVSTVLTSKYRZ 143

```

```

#####
# Program: water
# Rundate: Fri 18 Apr 2025 03:06:59
# Commandline: water
#   -asequence /corral4/main/objects/a/8/b/dataset_a8b599ae-5157-43f9-ae0c-84067cb25e21.dat
#   -bsequence /corral4/main/objects/f/6/d/dataset_f6da880b-4583-4808-8e39-197b1c8fcl1f6.dat
#   -outfile /corral4/main/jobs/066/907/66907251/outputs/dataset_3860bc43-86ff-46a9-881a-56ba3c05513d.dat
#   -gapopen 10.0
#   -gapextend 0.5
#   -brief yes
#   -aformat3 srs

```

```

# -auto
# Align_format: srs
# Report_file:
/corral4/main/jobs/066/907/66907251/outputs/dataset_3860bc43-86ff-46a9-
881a-56ba3c05513d.dat
#####

#=====
#
# Aligned_sequences: 2
# 1: NM_000517.6_hg19
# 2: NM_000518.5_hg19
# Matrix: EBLOSUM62
# Gap_penalty: 10.0
# Extend_penalty: 0.5
#
# Length: 147
# Identity:      64/147 (43.5%)
# Similarity:    89/147 (60.5%)
# Gaps:          8/147 ( 5.4%)
# Score: 297.5
#
#
#=====

NM_000517.6_h      3 LSPADKTNVKAAWGKVGHAHAGEYGAEALERMFSLFPTTKTYFPHF-DLS-
50
NM_000518.5_h      4 LTPEEKSAVTALWGKV--NVDEVGGEALGRLLVVYPWTQRFFESFGDLST
51

NM_000517.6_h      51 ----HGSAQVKGHGKKVADALTNAVAHVDDMPNALSALSDLHAHKLRVDP
96
NM_000518.5_h      52 PDAVMGNPKVKAHGKKVLGAFSDGLAHLNKGTFATLSELHCDKLHVDP
101

NM_000517.6_h      97 VNFKLLSHCLLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTSKYRZ      143
NM_000518.5_h     102 ENFRLLGNVLVLCVLAHFFGKEFTPPVQAAYQKVVAGVANALAHKYHZ      148

#=====
#
# Aligned_sequences: 2
# 1: NM_000517.6_hg19
# 2: NM_000518.5_equCab2
# Matrix: EBLOSUM62
# Gap_penalty: 10.0
# Extend_penalty: 0.5
#
# Length: 147
# Identity:      63/147 (42.9%)
# Similarity:    85/147 (57.8%)
# Gaps:          8/147 ( 5.4%)
# Score: 281.5
#
#
#=====

NM_000517.6_h      3 LSPADKTNVKAAWGKVGHAHAGEYGAEALERMFSLFPTTKTYFPHF-DLSH
51
NM_000518.5_e      4 LSGEEKAAVLALWDKVNEE--EVGGEALGRLLVVYPWTQRFFDSFGDLSN
51

```

NM_000517.6_h	52	-----GSAQVKGHGKKVADALTNAVAHVDDMPNALSALS	DLHAHKLRVDP
96			
NM_000518.5_e	52	PGAVMGNPVKVKAHGKKVLH	SFGEGVHHLDNLKGTFAALSELHCDKLHVDP
101			

NM_000517.6_h	97	VNFKLLSHCLLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTSKYRZ	143
NM_000518.5_e	102	ENFRLLGNVLVVVLARHFGKDFTPELQASYQKVAGVANALAHKYHZ	148

#-----
#-----